Week 9 : Graph BFS and DFS Objectives

Upon completion of this graph unit, students will be able to understand and explain the following sub topics :

- What Is Graph?
- Connected Graphs
- Paths
- Graph types BFS and DFS etc
- Graph Implementations

Graphs

- A road map is a graph
- Definition of a graph:
  - A collection of distinct vertices a
    distinct edges
- In the map of Figure 29-1
  - The circles are vertices or nodes
  - The lines are called edges
- A subgraph is a portion of a graph that is itself a graph

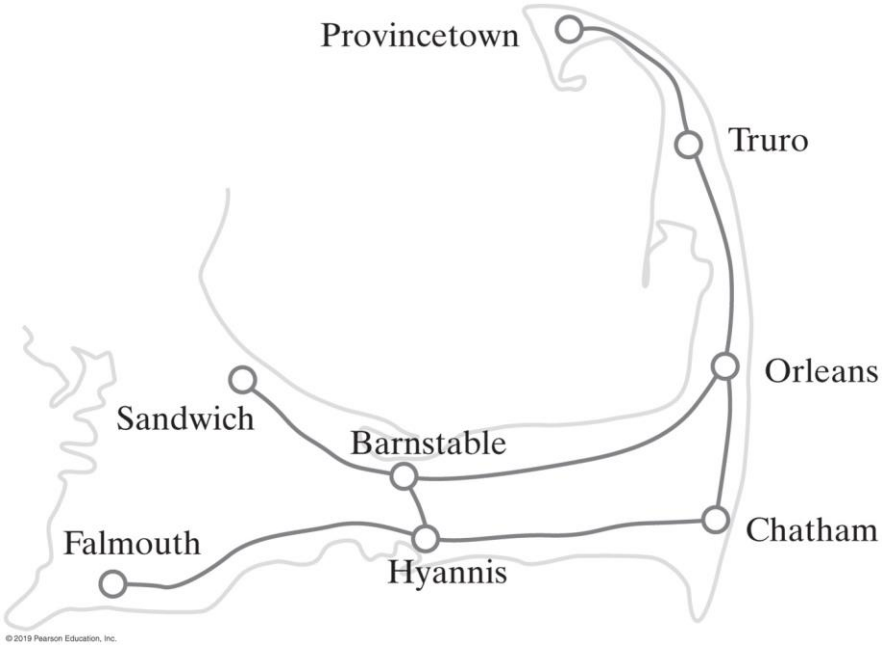


FIGURE 29-1 A portion of a road map

Connected Graphs

- A graph that has a path between every pair of distinct vertices is connected
- A complete graph has an edge between every pair of distinct vertices.
- Undirected graphs can be
  - Connected
  - Complete or
  - Disconnected

Graphs

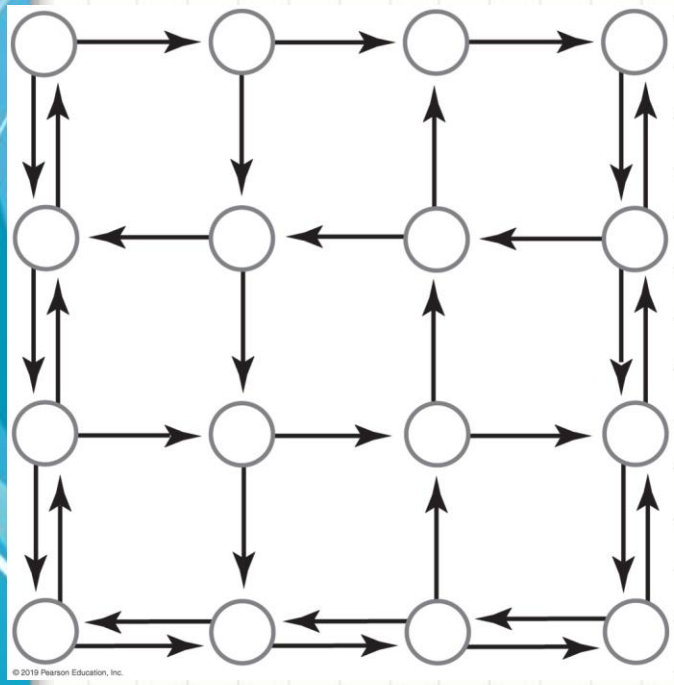
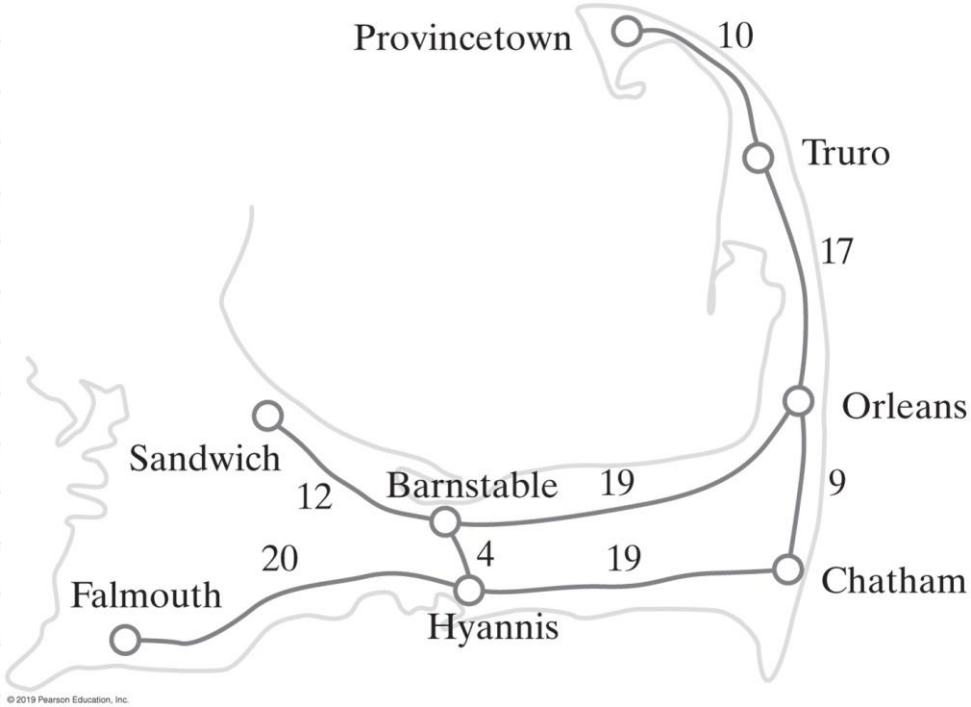


FIGURE 29-2 A directed graph representing a portion of a city's street map



- FIGURE 29-3 A weighted graph

Paths

- A path between two vertices in a graph is a sequence of edges
- A path in a directed graph must consider the direction of the edges
  - Called a directed path
- The length of a path is the number of edges that it comprises.
- A cycle is a path that begins and ends at the same vertex

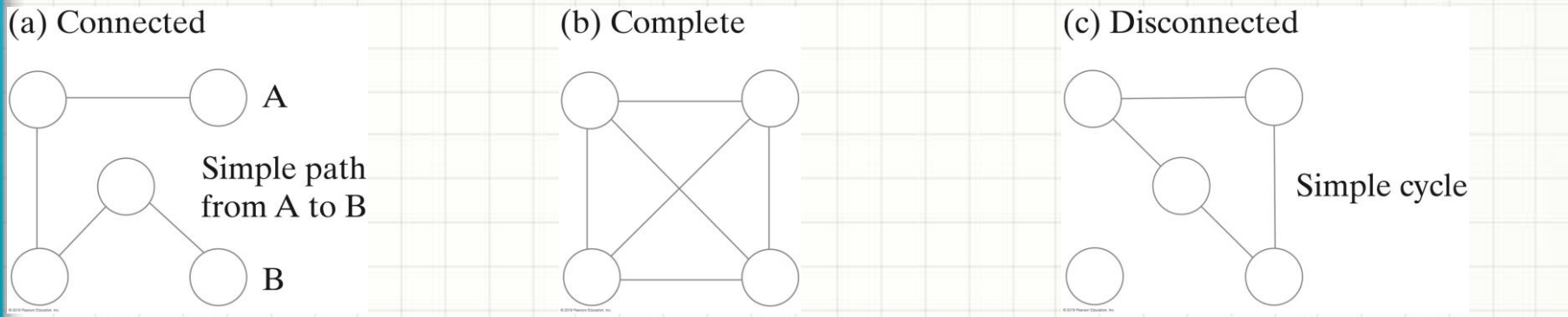
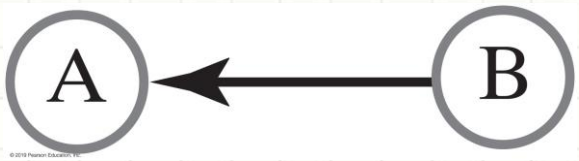
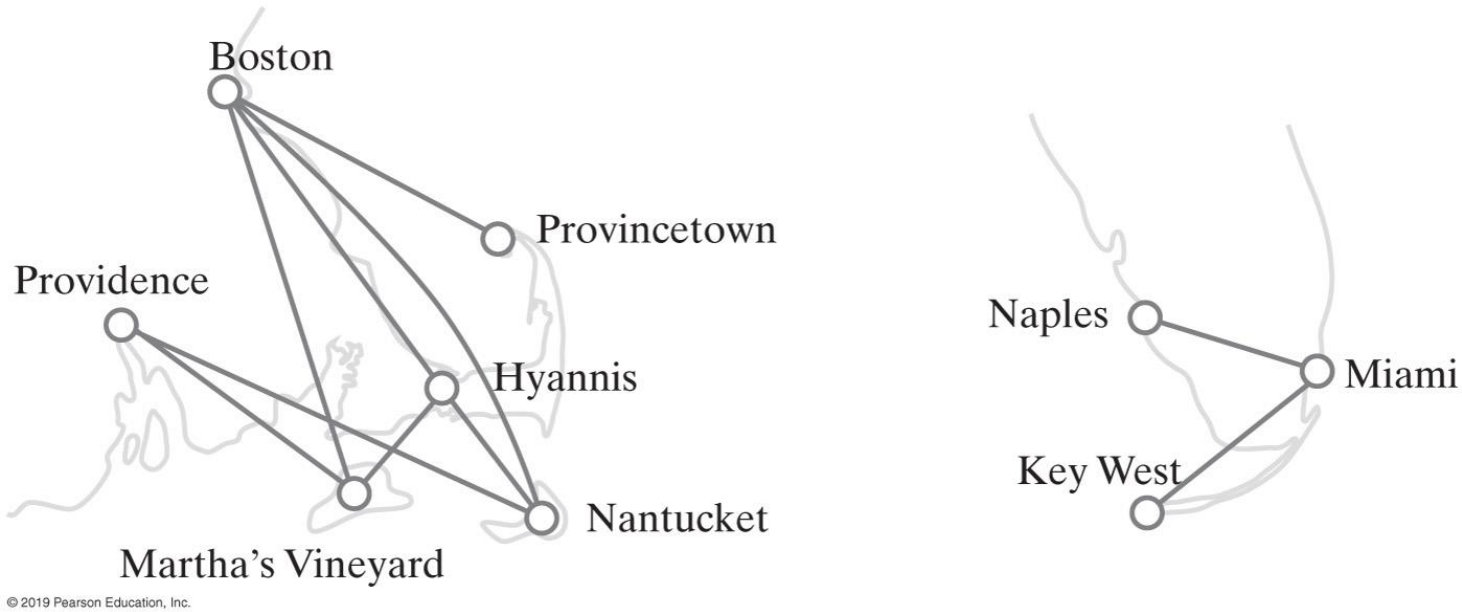


FIGURE 29-4 Undirected graphs

Directed Graphs

- FIGURE 29-5 Vertex A is adjacent to vertex B, but B is not adjacent to A

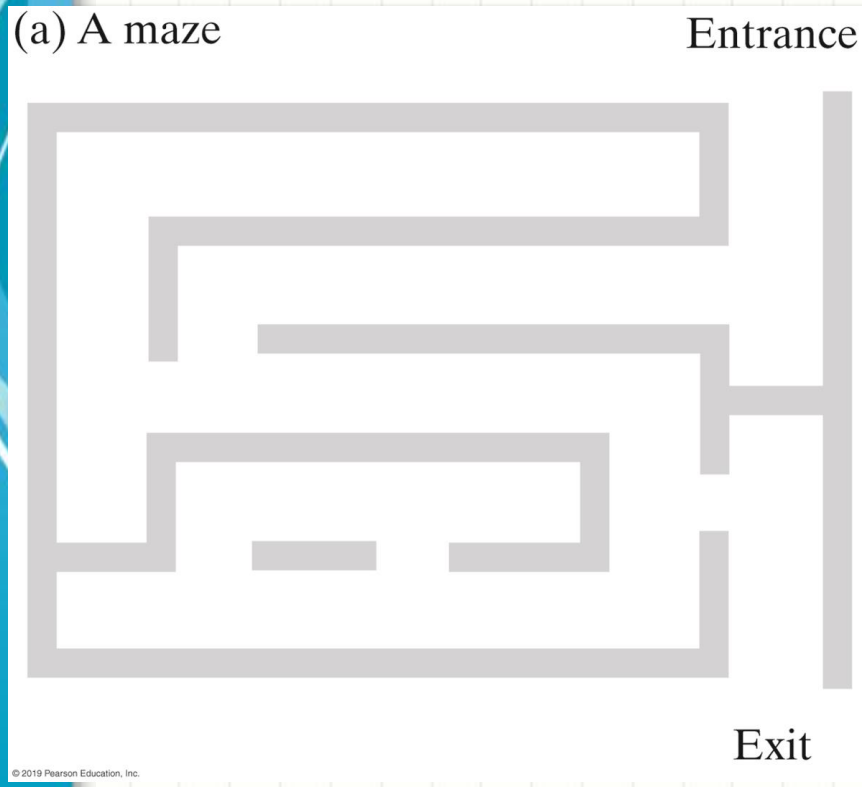


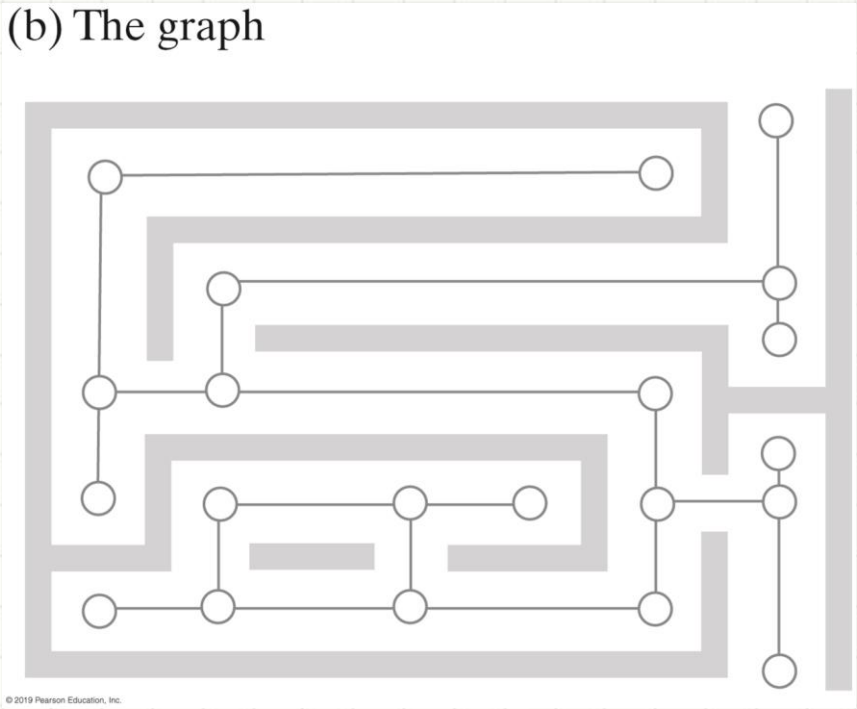


Airline routes

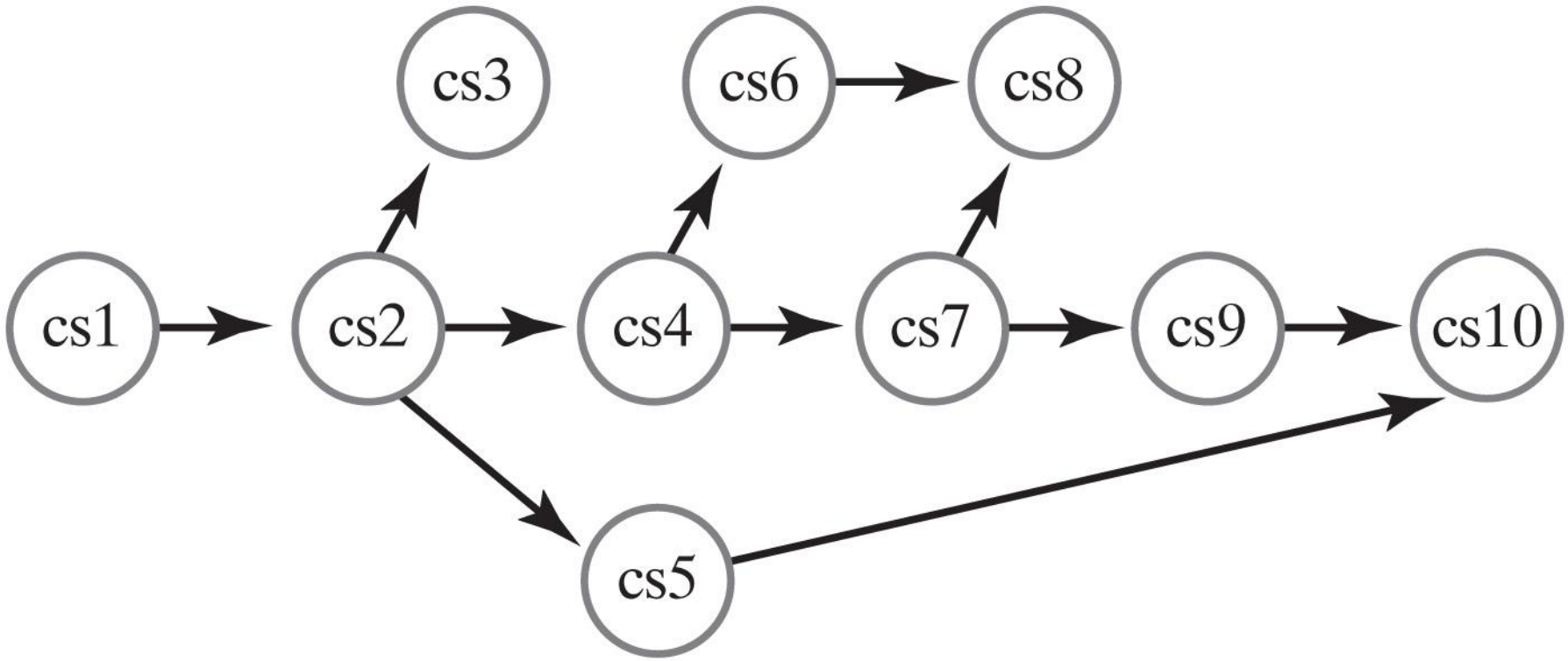
Graphs

- A maze and its representation as a graph





Directed Graphs



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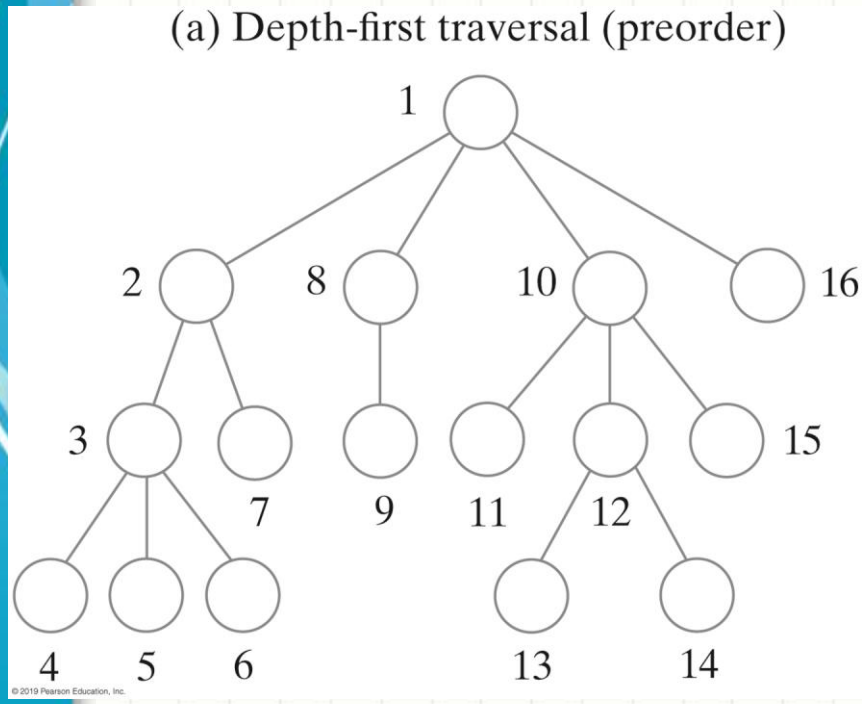
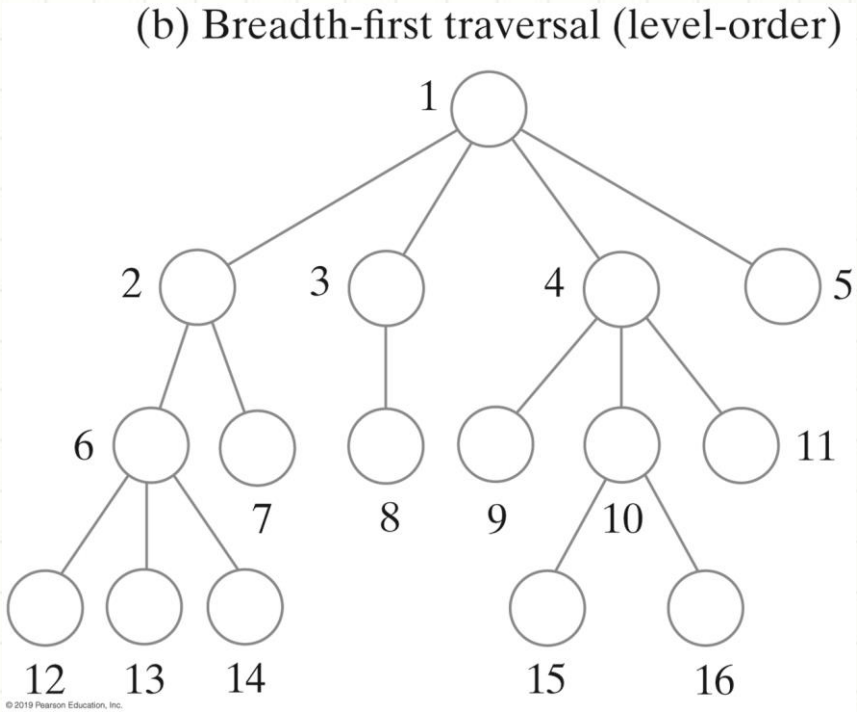
- The prerequisite structure for a selection of courses as a directed graph without cycles

Graph Traversal





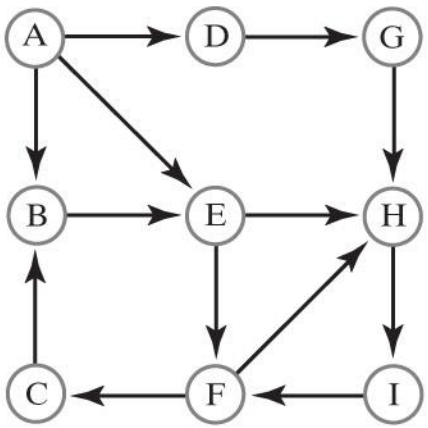
- The visitation order of two traversals

Breadth-First Traversal

```
Algorithm getBreadthFirstTraversal(originVertex)
traversalOrder = a new queue for the resulting traversal order
vertexQueue = a new queue to hold vertices as they are visited
Mark originVertex as visited
traversalOrder.enqueue(originVertex)
vertexQueue.enqueue(originVertex)
while (!vertexQueue.isEmpty())
{
frontVertex = vertexQueue.dequeue()
        while (frontVertex has a neighbor)
        {
                nextNeighbor = next neighbor of frontVertex
                if (nextNeighbor is not visited)
                {
                        Mark nextNeighbor as visited
                        traversalOrder.enqueue(nextNeighbor)
                        vertexQueue.enqueue(nextNeighbor)
                }
        }
}
return traversalOrder
```

- Algorithm that performs a breadth-first traversal of a nonempty graph beginning at a given vertex

Breadth-First Traversal



frontVertex	nextNeighbor	Visited vertex	vertexQueue (front to back)	traversalOrder
		A	A	A
A			*empty*	
	B	B	B	A B
	D	D	B D	A B D
	E	E	B D E	A B D E
B			D E	
D			E	
	G	G	E G	A B D E G
E			G	
	F	F	G F	A B D E G F
	H	H	G F H	A B D E G F H
G			F H	
F			H	
	C	C	H C	A B D E G F H C
H			C	
	I	I	C I	A B D E G F H C I
C			I	
I			*empty*	

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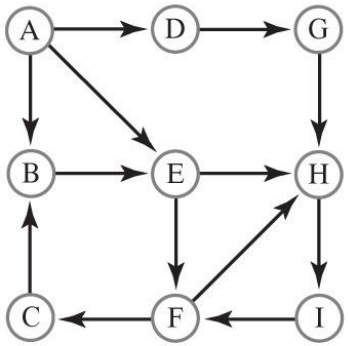
- A trace of a breadth-first traversal beginning at vertex A of a directed graph

Depth-First Traversal

```
Algorithm getDepthFirstTraversal(originVertex)
traversalOrder = a new queue for the resulting traversal order
vertexStack = a new stack to hold vertices as they are visited
Mark originVertex as visited
traversalOrder.enqueue(originVertex)
vertexStack.push(originVertex)
while  (!vertexStack.isEmpty())
{
        topVertex = vertexStack.peek()
        if (topVertex has an unvisited neighbor)
        {
                nextNeighbor = next unvisited neighbor of
            topVertex
                Mark nextNeighbor as visited
                traversalOrder.enqueue(nextNeighbor)
                vertexStack.push(nextNeighbor)
        }
        else // All neighbors are visited
                vertexStack.pop()
}
return traversalOrder
```

- Algorithm that performs a depth-first traversal of a nonempty graph, beginning at a given vertex

Depth-First Traversal



topVertex	nextNeighbor	Visited vertex	vertexStack (top to bottom)	traversalOrder (front to back)
		A	A	A
A			A	
	B	B	B A	A B
B			B A	
	E	E	E B A	A B E
E			E B A	
	F	F	F E B A	A B E F
F			F E B A	
	C	C	C F E B A	A B E F C
C			F E B A	
F			F E B A	
	H	H	H F E B A	A B E F C H
H			H F E B A	
	I	I	I H F E B A	A B E F C H I
I			H F E B A	
H			F E B A	
F			E B A	
E			B A	
B			A	
A			A	
	D	D	D A	A B E F C H I D
D			D A	
	G		G D A	A B E F C H I D G
G			D A	
D			A	
A			*empty*	

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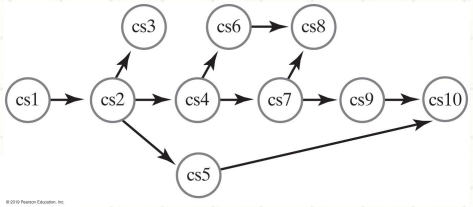
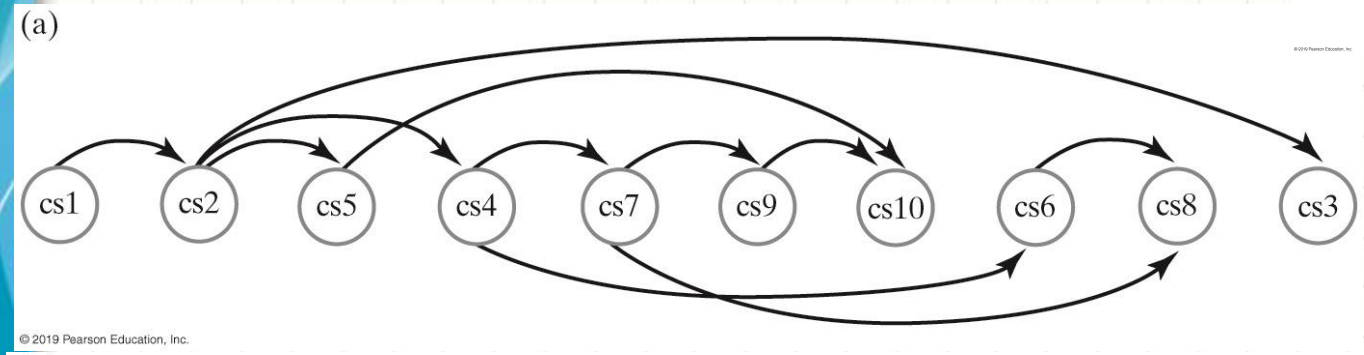
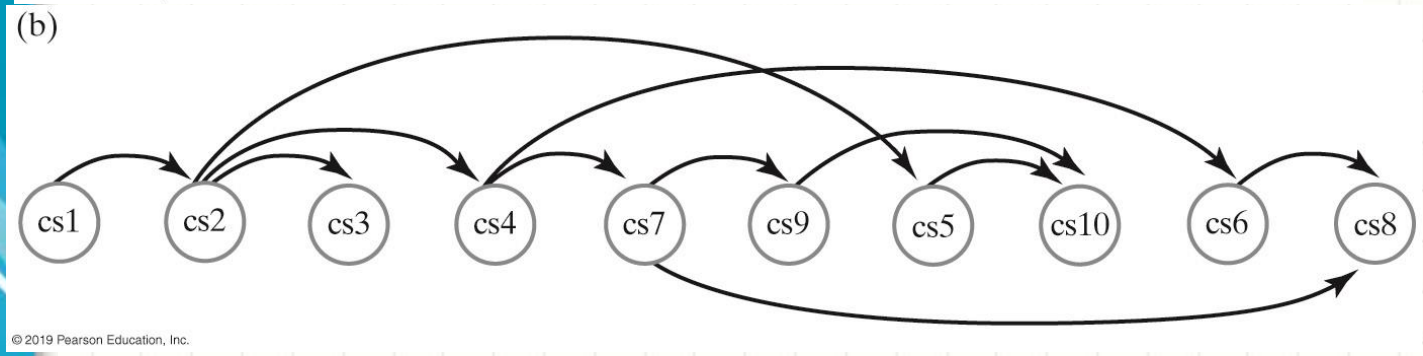
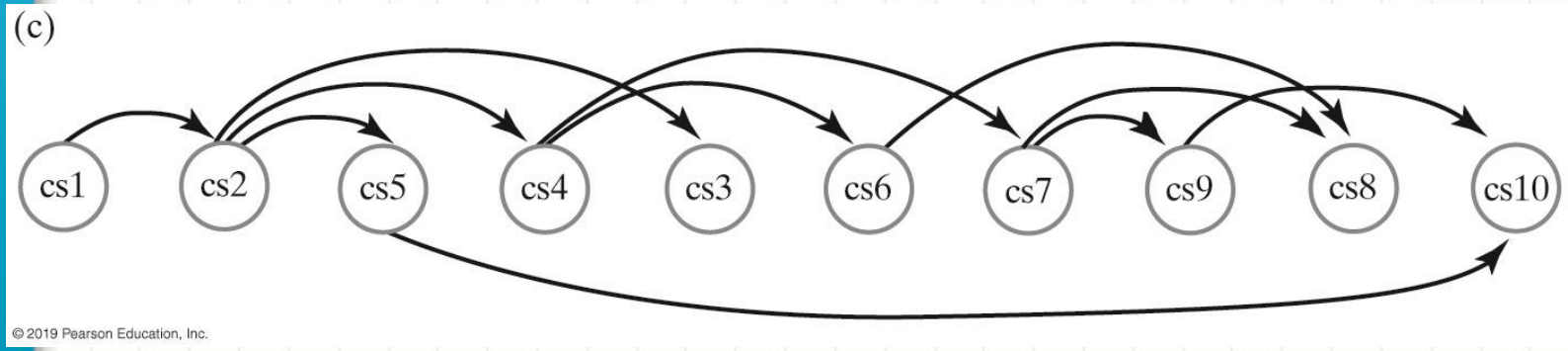
- A trace of a depth-first traversal beginning at vertex A of a directed graph

Topological Ordering









- Three topological orders for the graph in above Figure

Topological Ordering

In a topological order of the vertices in a directed graph without cycles, vertex a precedes vertex b whenever a directed edge exists from a to b.

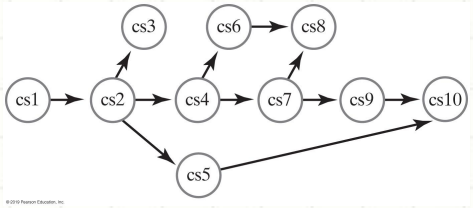
- FIGURE 29-13 An impossible prerequisite structure for three courses, as a directed graph with a cycle

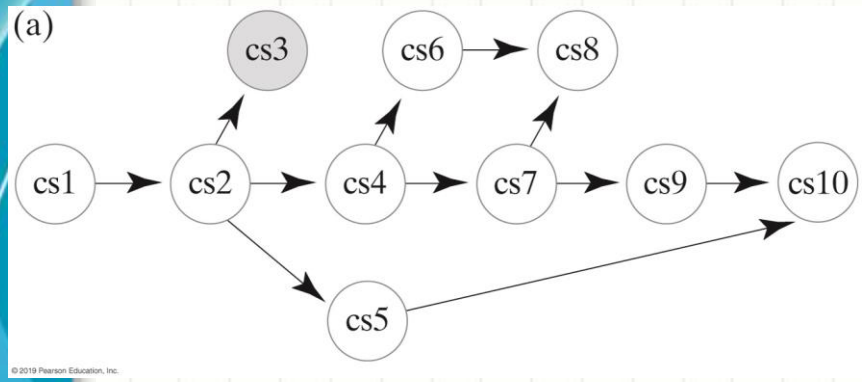
Topological Order

```
Algorithm getTopologicalOrder()
vertexStack = a new stack to hold vertices as they are visited
numberOfVertices = number of vertices in the graph
for (counter = 1 to numberOfVertices)
{
        nextVertex = an unvisited vertex whose neighbors, if any, are all visited
        Mark nextVertex as visited
        vertexStack.push(nextVertex)
}
return vertexStack
```

- An algorithm that describes a topological sort

Topological Ordering (Part 1)





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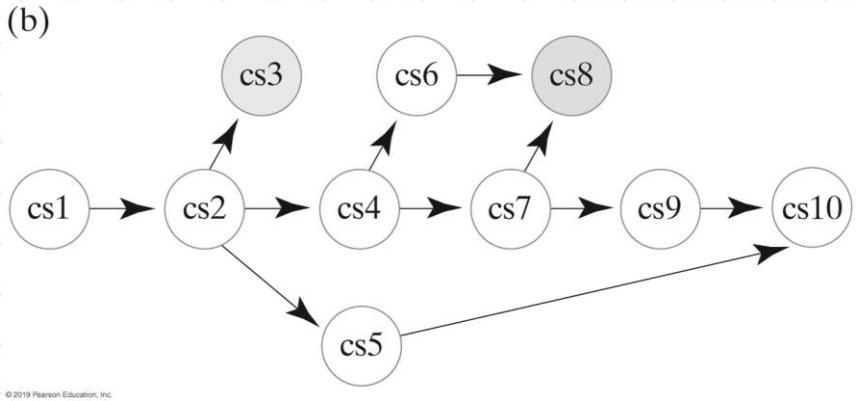
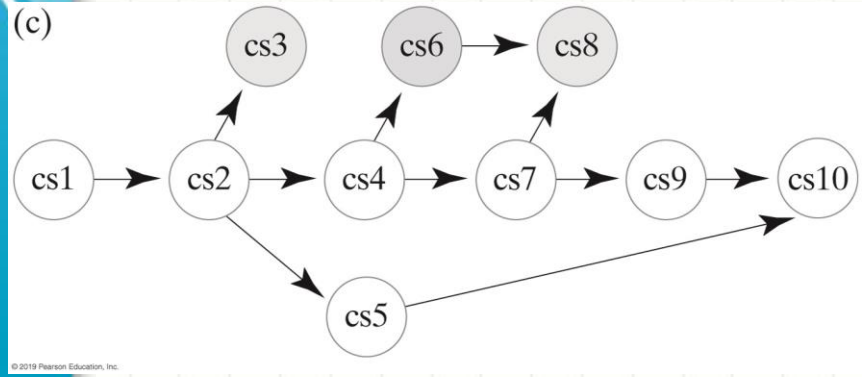
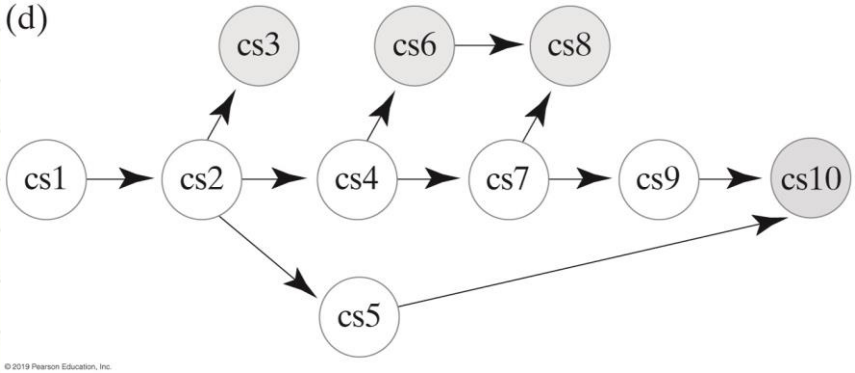
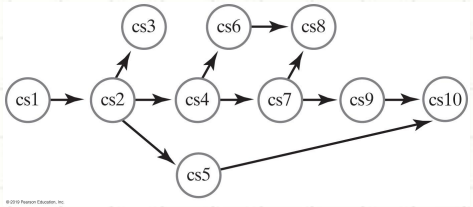
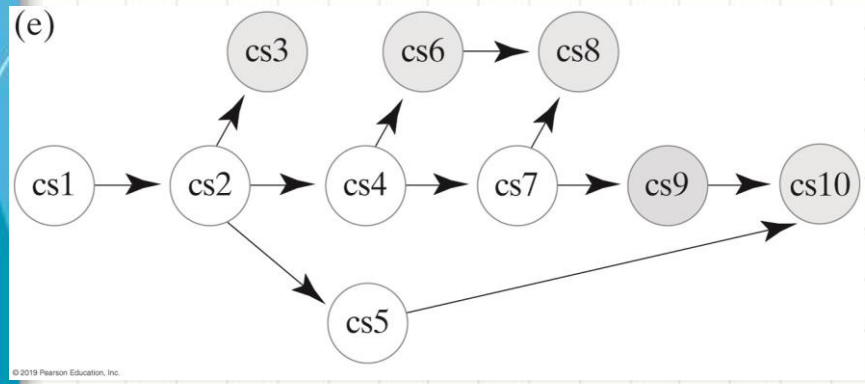


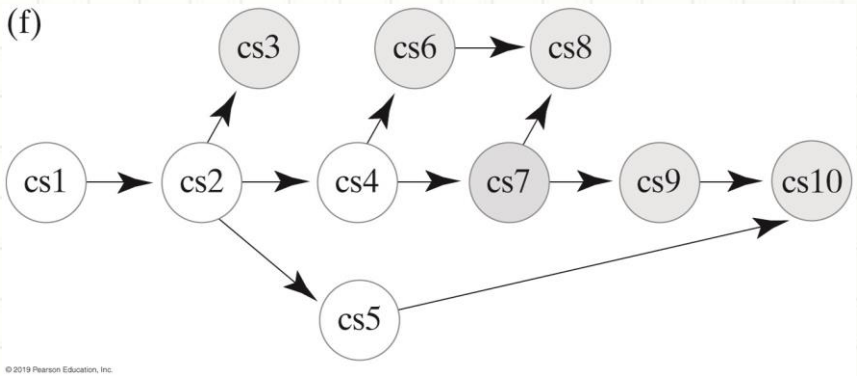


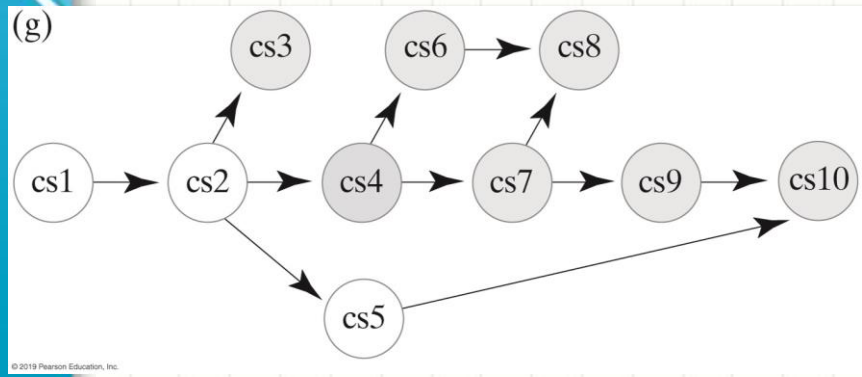


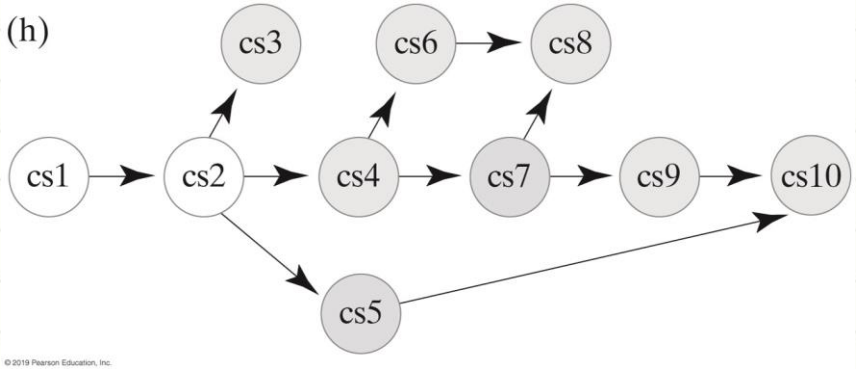
Topological Ordering (Part 2)











- Finding a topological order for the graph in Figure

Topological Ordering (Part 3)

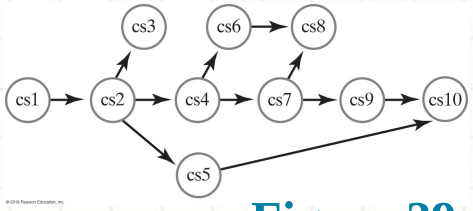
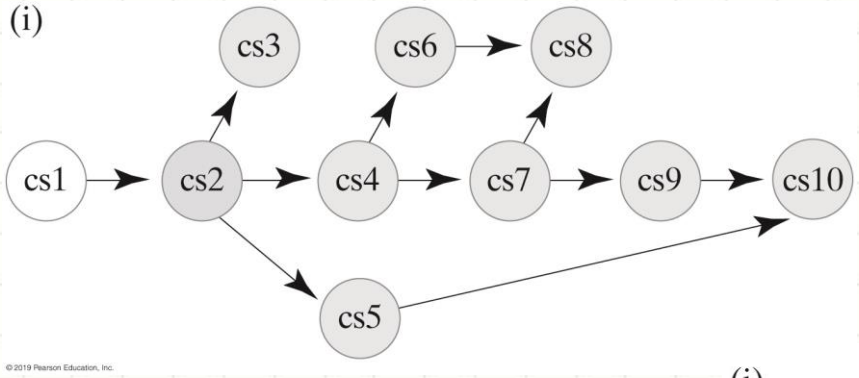
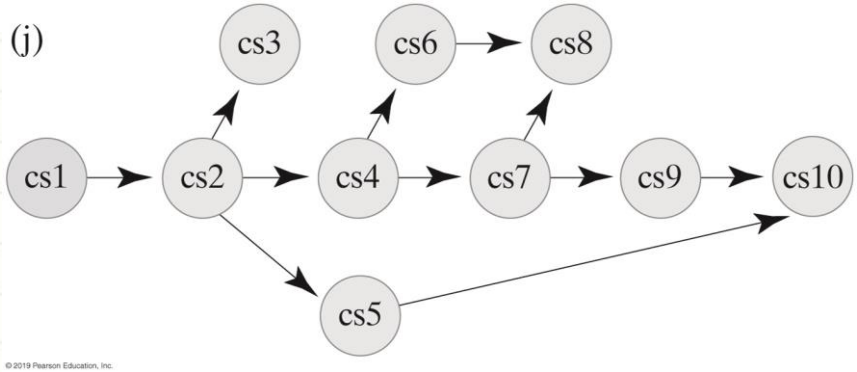


Figure 29-8

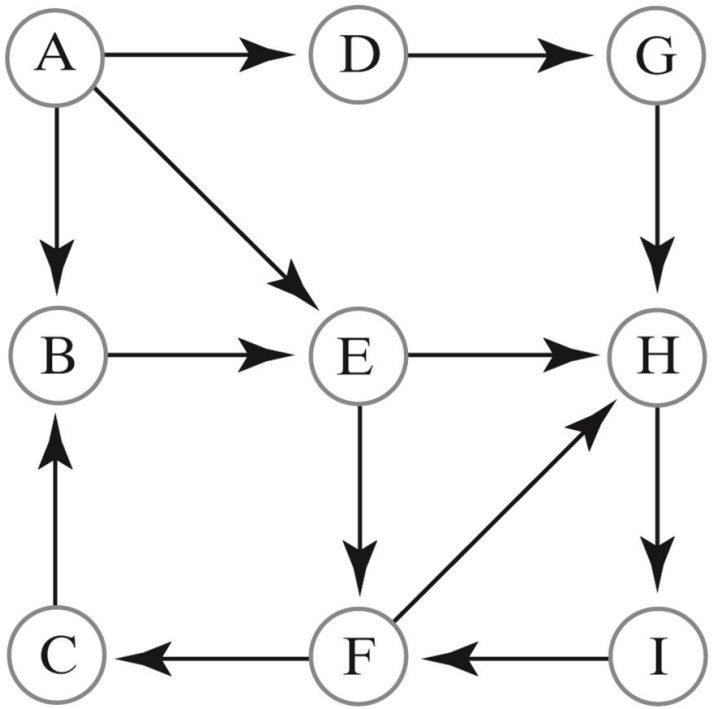




- Finding a topological order for the graph in Figure 29-8

Shortest Path in Unweighted Graph

(a) An unweighted graph



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(b) Possible paths through the graph

A → B → E → F → H
A → B → E → H
A → D → G → H
A → E → F → H
A → E → H

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- An unweighted graph and the possible paths from vertex A to vertex H

Shortest Path in an Unweighted Graph (Part 1)

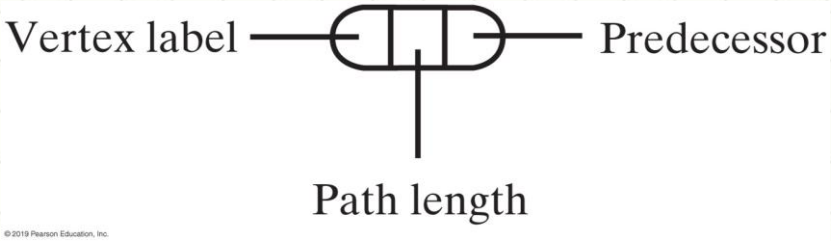
```
Algorithm getShortestPath(originVertex, endVertex, path)
done = false
vertexQueue = a new queue to hold vertices as they are visited
Mark originVertex as visited
vertexQueue.enqueue(originVertex)
while (!done && !vertexQueue.isEmpty())
{
    frontVertex = vertexQueue.dequeue()
    while (!done && frontVertex has a neighbor)
    {
        nextNeighbor = next neighbor of frontVertex
        if (nextNeighbor is not visited)
        {
            Mark nextNeighbor as visited
            Set the length of the path to nextNeighbor to 1 + length of path to frontVertex
            Set the predecessor of nextNeighbor to frontVertex
            vertexQueue.enqueue(nextNeighbor)
        }
        if (nextNeighbor equals endVertex)
            done = true
    }
}
```

- Algorithm to find shortest path

Shortest Path in an Unweighted Graph (Part 2)

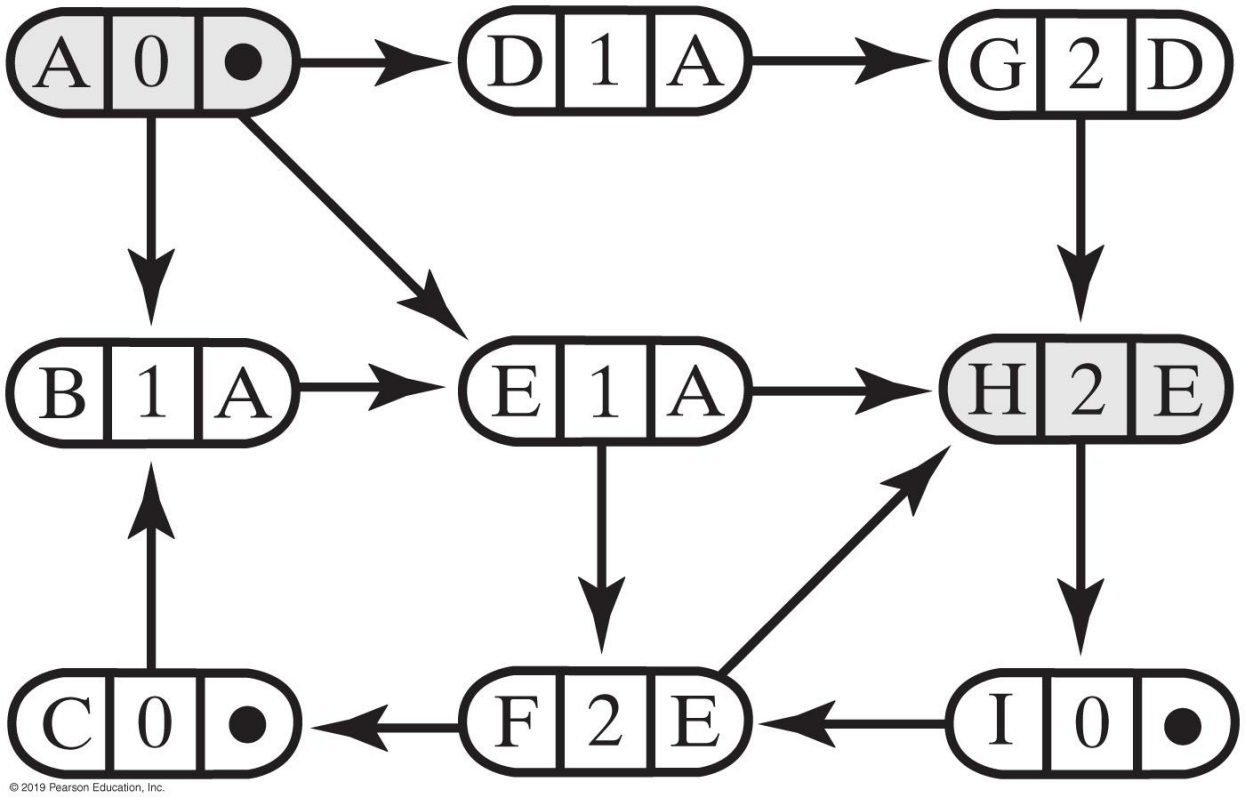
```
// Traversal ends; construct shortest path
pathLength = length of path to endVertex
path.push(endVertex)
vertex = endVertex
while (vertex has a predecessor)
{
        vertex = predecessor of vertex
        path.push(vertex)
}
return pathLength
```

- Algorithm to find shortest path



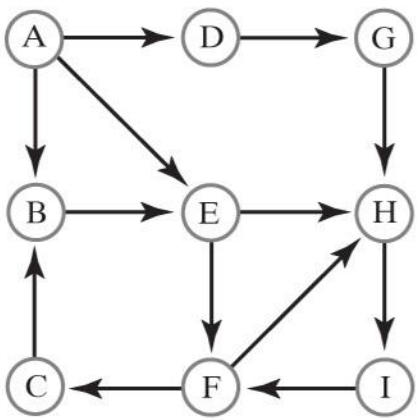
The data in a vertex

Shortest Path in a Weighted Graph



- The graph in previous Figure after the shortest-path algorithm has traversed from vertex A to vertex H

Shortest Path in a Weighted Graph



frontVertex	nextNeighbor	Visited vertex	vertexQueue (front to back)
		A	A 0 •
A 0 •			*empty*
	B	B	B 1 A
	D	D	B 1 A, D 1 A
	E	E	B 1 A, D 1 A, E 1 A
B 1 A			D 1 A, E 1 A
D 1 A			E 1 A
	G	G	E 1 A, G 2 D
E 1 A			G 2 D
	F	F	G 2 D, F 2 E
	H	H	G 2 D, F 2 E, H 2 E

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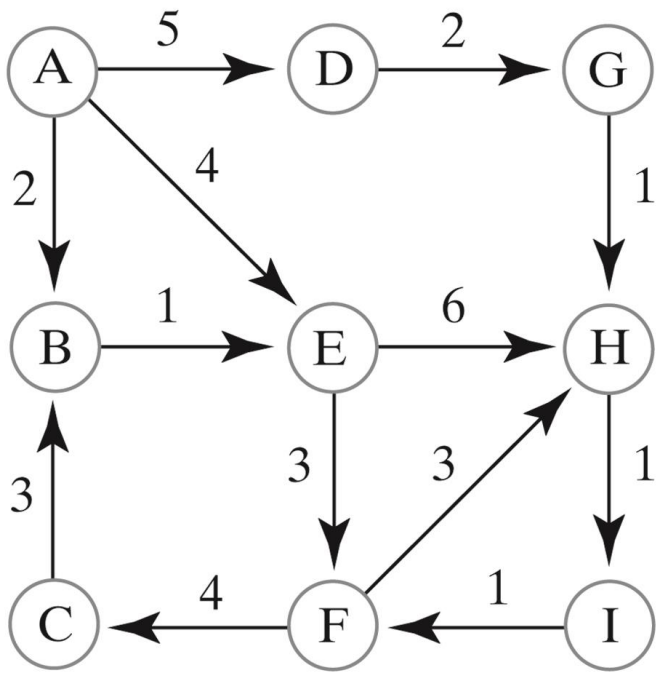
- FIGURE 29-18 A trace of the traversal in the algorithm to find the shortest path from vertex A to vertex H in an unweighted graph

Shortest Path in a Weighted Graph

(a) A weighted graph



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h and the

(b) Possible paths and their weights

Path	Weight
A → B → E → F → H	9
A → B → E → H	9
A → D → G → H	8
A → E → F → H	10
A → E → H	10

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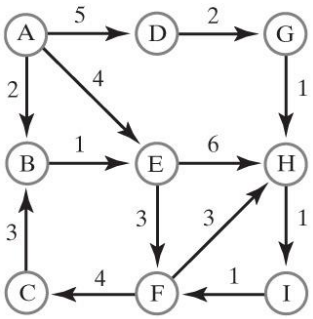
Shortest Path in a Weighted Graph



frontVertex	Visited vertex	nextNeighbor	Priority queue (front to back)
			A 0 •
A 0 •	A		*empty*
		B	B 2 A
		D	B 2 A, D 5 A
		E	B 2 A, E 4 A, D 5 A
B 2 A	B		E 4 A, D 5 A
		E	E 3 B, E 4 A, D 5 A
E 3 B	E		E 4 A, D 5 A
		F	E 4 A, D 5 A, F 6 E
		H	E 4 A, D 5 A, F 6 E, H 9 E
			D 5 A, F 6 E, H 9 E
D 5 A	D		F 6 E, H 9 E
		G	F 6 E, G 7 D, H 9 E
F 6 E	F		G 7 D, H 9 E
		H	G 7 D, H 9 E, H 9 F
		C	G 7 D, H 9 E, H 9 F, C 10 F
G 7 D	G		H 9 E, H 9 F, C 10 F
		H	H 8 G, H 9 E, H 9 F, C 10 F
H 8 G	H		H 9 E, H 9 F, C 10 F

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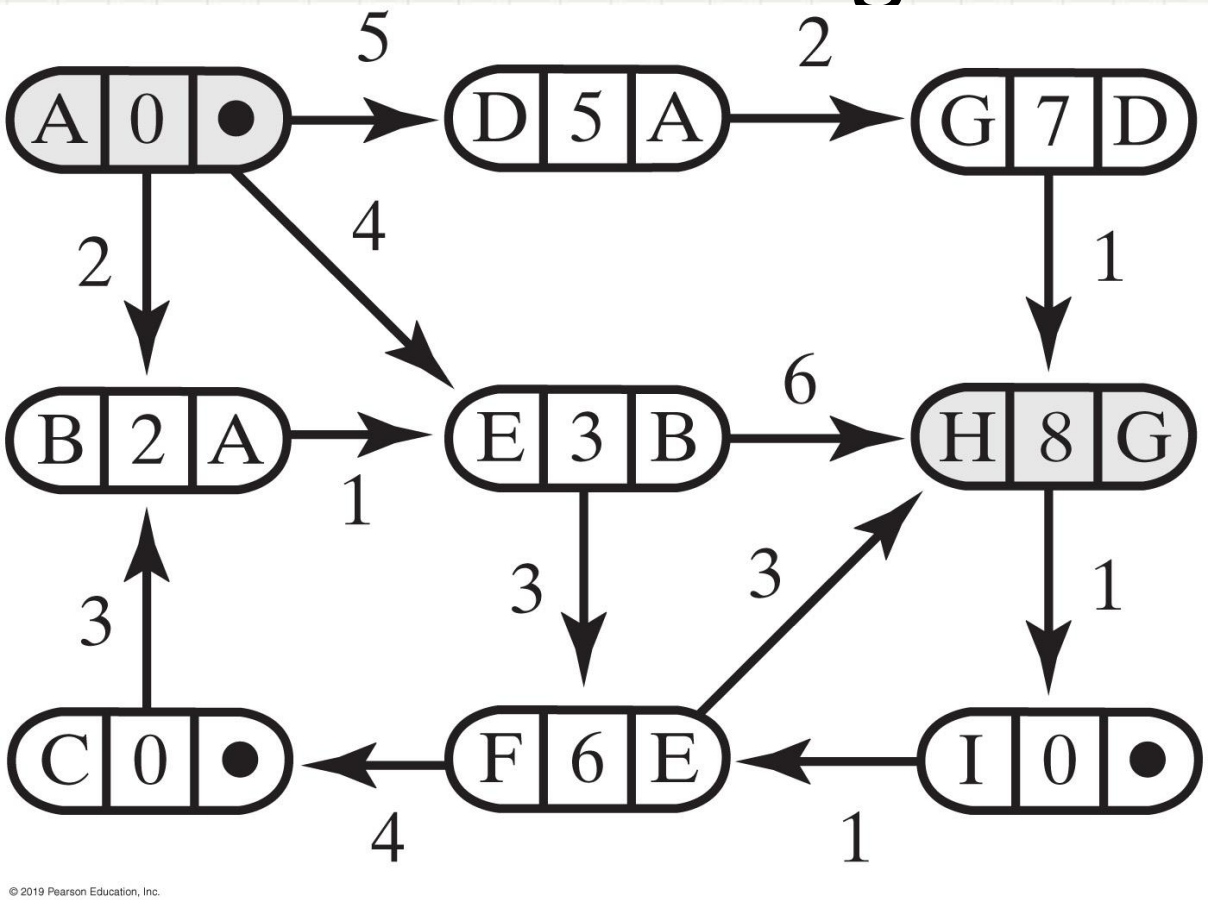
- A trace of the traversal in the algorithm to find the cheapest path from vertex A to vertex H in a weighted graph

Shortest Path in a Weighted Graph



- FIGURE 29-21 The graph in Figure 29-19a after finding the cheapest path from vertex A to vertex H

The Shortest Path in a Weighted Graph (Part 1)

- Pseudocode for shortest path (weighted)

```
Algorithm getCheapestPath(originVertex, endVertex, path)
done = false
priorityQueue = a new priority queue
priorityQueue.add(new EntryPQ(originVertex, 0, null))
while (!done && !priorityQueue.isEmpty())
{
        frontEntry = priorityQueue.remove()
        frontVertex = vertex in frontEntry
        if (frontVertex is not visited)
        {
                Mark frontVertex as visited
                Set the cost of the path to frontVertex to the cost recorded in fro
                Set the predecessor of frontVertex to the predecessor recorded i
                if (frontVertex equals endVertex)
                        done = true
                else
                {
```

The Shortest Path in a Weighted Graph (Part 2)

- Pseudocode for shortest path (weighted)

```
                    while (frontVertex has a neighbor)
                    {
                        nextNeighbor = next neighbor of fro
                        weightOfEdgeToNeighbor = weight
                        if (nextNeighbor is not visited)
                        {
            nextCost = weightOfEdgeToNeighbor + cost of path to frontVertex
            priorityQueue.add(new EntryPQ(nextNeighbor, nextCost,
                                                    frontVertex))
                        }
                    }
                }
            }
}
```

The Shortest Path in a Weighted Graph (Part 3)

- Pseudocode for shortest path (weighted)

```
// Traversal ends; construct cheapest path
pathCost = cost of path to endVertex
path.push(endVertex)
vertex = endVertex
while (vertex has a predecessor)
{
        vertex = predecessor of vertex
        path.push(vertex)
}
return pathCost
```

Java Interfaces for the ADT Graph (Part 1)

```
/** An interface of methods providing basic operations for directed
   and undirected graphs that are either weighted or unweighted. */
public interface BasicGraphInterface<T>
{
   /** Adds a given vertex to this graph.
      @param vertexLabel  An object that labels the new vertex and is
                          distinct from the labels of current vertices.
      @return  True if the vertex is added, or false if not. */
   public boolean addVertex(T vertexLabel);

   /** Adds a weighted edge between two given distinct vertices that
      are currently in this graph. The desired edge must not already
      be in the graph. In a directed graph, the edge points toward
      the second vertex given.
      @param begin  An object that labels the origin vertex of the edge.
      @param end    An object, distinct from begin, that labels the end
                    vertex of the edge.
      @param edgeWeight  The real value of the edge's weight.
      @return  True if the edge is added, or false if not. */
   public boolean addEdge(T begin, T end, double edgeWeight);
```

- LISTING 29-1 An interface of basic graph operations

Java Interfaces for the ADT Graph (Part 2)

```
/** Adds an unweighted edge between two given distinct vertices
    that are currently in this graph. The desired edge must not
    already be in the graph. In a directed graph, the edge points
    toward the second vertex given.
    @param begin  An object that labels the origin vertex of the edge.
    @param end    An object, distinct from begin, that labels the end
                  vertex of the edge.
    @return  True if the edge is added, or false if not. */
public boolean addEdge(T begin, T end);

/** Sees whether an edge exists between two given vertices.
    @param begin  An object that labels the origin vertex of the edge.
    @param end    An object that labels the end vertex of the edge.
    @return  True if an edge exists. */
public boolean hasEdge(T begin, T end);

/** Sees whether this graph is empty.
    @return  True if the graph is empty. */
public boolean isEmpty();
```

- LISTING 29-1 An interface of basic graph operations

Java Interfaces for the ADT Graph (Part 3)

- LISTING 29-1 An interface of basic graph operations

```
/** Gets the number of vertices in this graph.
    @return  The number of vertices in the graph. */
public int getNumberOfVertices();

/** Gets the number of edges in this graph.
   @return  The number of edges in the graph. */
public int getNumberOfEdges();

/** Removes all vertices and edges from this graph
                                resulting in an empty graph. */
public void clear();
} // end BasicGraphInterface
```

Java Interfaces for the ADT Graph

- FIGURE 29-22 A portion of the flight map in Figure 29-6

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Java Interfaces for the ADT Graph (Part 1)

```
/** An interface of methods that process an existing graph. */
public interface GraphAlgorithmsInterface<T>
{
   /** Performs a breadth-first traversal of this graph.
      @param origin  An object that labels the origin vertex of the traversal.
      @return  A queue of labels of the vertices in the traversal, with
               the label of the origin vertex at the queue's front. */
   public QueueInterface<T> getBreadthFirstTraversal(T origin);

   /** Performs a depth-first traversal of this graph.
      @param origin  An object that labels the origin vertex of the traversal.
      @return  A queue of labels of the vertices in the traversal, with
               the label of the origin vertex at the queue's front. */
   public QueueInterface<T> getDepthFirstTraversal(T origin);

   /** Performs a topological sort of the vertices in this graph without cycles.
      @return  A stack of vertex labels in topological order, beginning
               with the stack's top. */
   public StackInterface<T> getTopologicalOrder();
```

- LISTING 29-2 An interface of operations on an existing graph

Java Interfaces for the ADT Graph (Part 2)

- LISTING 29-2 An interface of operations on an existing graph

```
/** Finds the shortest-length path between two given vertices in this graph.
    @param begin  An object that labels the path's origin vertex.
    @param end    An object that labels the path's destination vertex.
    @param path   A stack of labels that is empty initially;
                  at the completion of the method, this stack contains
                  the labels of the vertices along the shortest path;
                  the label of the origin vertex is at the top, and
                  the label of the destination vertex is at the bottom
    @return  The length of the shortest path. */
public int getShortestPath(T begin, T end, StackInterface<T> path);

/** Finds the least-cost path between two given vertices in this graph.
    @param begin  An object that labels the path's origin vertex.
    @param end    An object that labels the path's destination vertex.
    @param path   A stack of labels that is empty initially;
                  at the completion of the method, this stack contains
                  the labels of the vertices along the cheapest path;
                  the label of the origin vertex is at the top, and
                  the label of the destination vertex is at the bottom
    @return  The cost of the cheapest path. */
```

Java Interfaces for the ADT Graph

- LISTING 29-3 An interface for the ADT graph

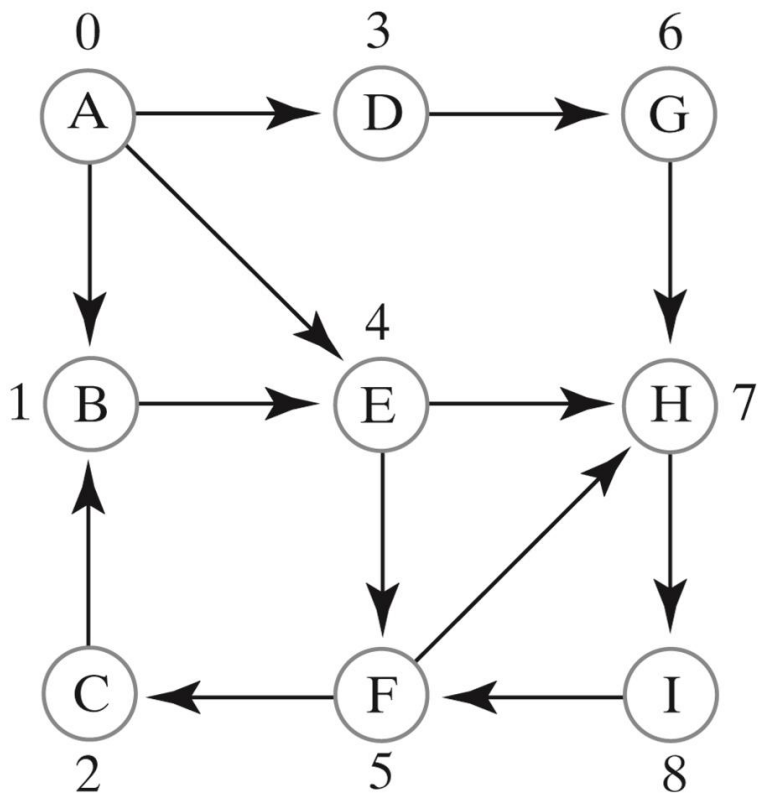
```
/** An interface of methods that create, manipulate, and process a graph. */
public interface GraphInterface<T> extends BasicGraphInterface<T>,
                                           GraphAlgorithmsInterface<T>
{
} // end GraphInterface
```

Overview of Two Implementations

- Two common implementations of the ADT graph
  - Array:
    - Typically a two-dimensional array called an *adjacency matrix*
  - List:
    - Referred to as an *adjacency list*.

FIGURE 30-1 An unweighted, directed graph and its adjacency matrix

(a) A graph



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(b) The graph's adjacency matrix

	A	B	C	D	E	F	G	H	I	
A		T		T	T					0
B					T					1
C		T								2
D							T			3
E						T		T		4
F			T					T		5
G								T		6
H									T	7
I						T				8
	0	1	2	3	4	5	6	7	8	

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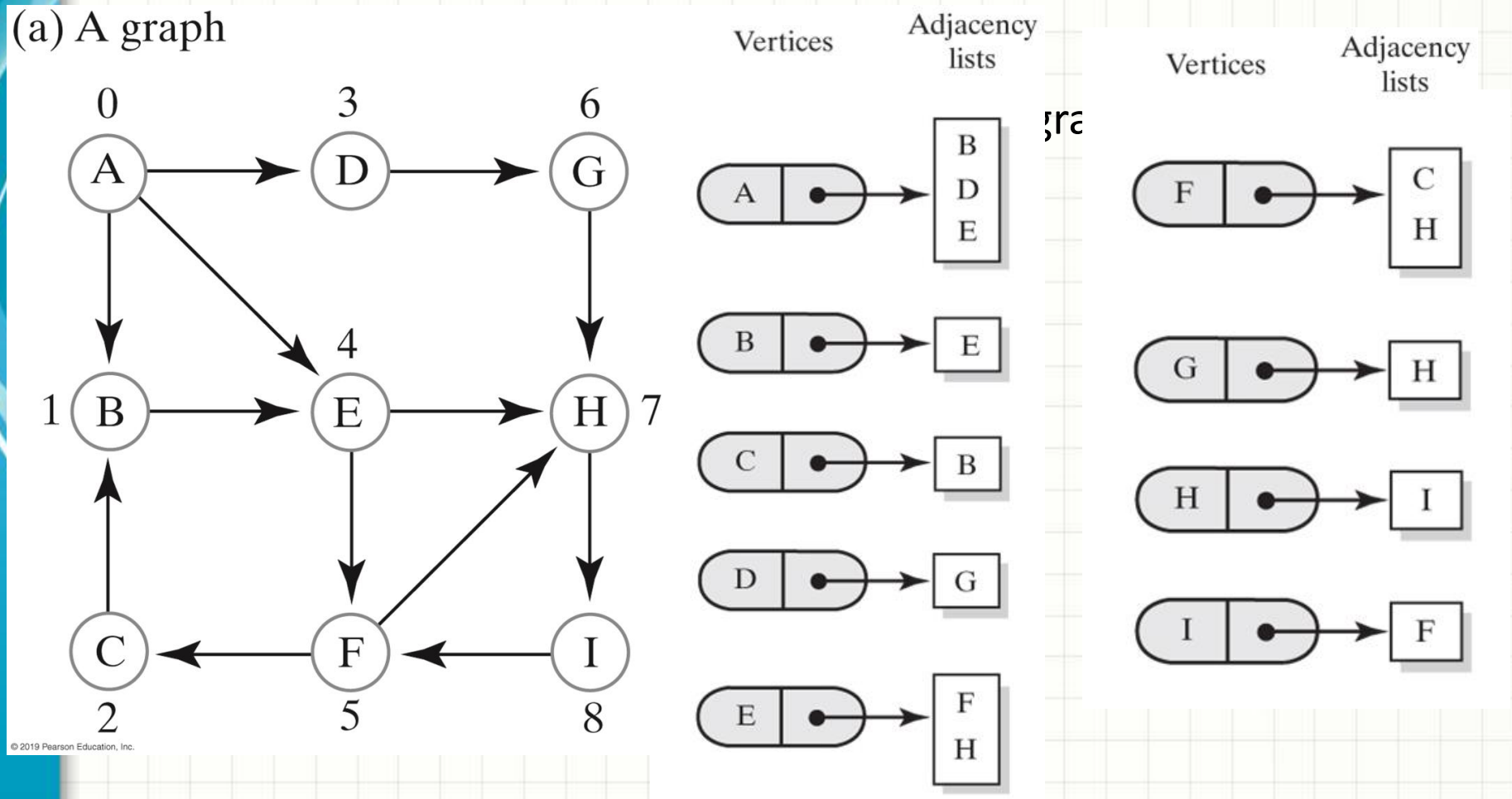
FIGURE 30-2 Adjacency lists for the directed graph in Figure 30-1a



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Vertices and Edges

- Specifying the Class **`Vertex`**
  - Identify vertices
  - Visit vertices
  - Adjacency list
  - Path operations

Vertices and Edges — Interface (Part 1)

```
/**  An interface for a vertex in a graph. */
public interface VertexInterface<T>
{
   /** Gets this vertex's label.
       @return  The object that labels the vertex. */
   public T getLabel();

   /** Marks this vertex as visited. */
   public void visit();

   /** Removes this vertex's visited mark. */
   public void unvisit();

   /** Sees whether the vertex is marked as visited.
       @return  True if the vertex is visited. */
   public boolean isVisited();

   /** Connects this vertex and a given vertex with a weighted edge.
       The two vertices cannot be the same, and must not already
       have this edge between them. In a directed graph, the edge
       points toward the given vertex.
       @param endVertex   A vertex in the graph that ends the edge.
       @param edgeWeight  A real-valued edge weight, if any.
       @return  True if the edge is added, or false if not. */
   public boolean connect(VertexInterface<T> endVertex, double edgeWeight);
```

- LISTING 30-1 An interface for the vertices in a graph

Vertices and Edges — Interface (Part 2)

```
/** Connects this vertex and a given vertex with an unweighted
    edge. The two vertices cannot be the same, and must not
    already have this edge between them. In a directed graph,
    the edge points toward the given vertex.
    @param endVertex   A vertex in the graph that ends the edge.
    @return  True if the edge is added, or false if not. */
public boolean connect(VertexInterface<T> endVertex);

/** Creates an iterator of this vertex's neighbors by following
    all edges that begin at this vertex.
    @return  An iterator of the neighboring vertices of this vertex. */
public Iterator<VertexInterface<T>> getNeighborIterator();

/** Creates an iterator of the weights of the edges to this
    vertex's neighbors.
 @return  An iterator of edge weights for edges to neighbors of this vertex. */
public Iterator<Double> getWeightIterator();

/** Sees whether this vertex has at least one neighbor.
    @return  True if the vertex has a neighbor. */
public boolean hasNeighbor();

/** Gets an unvisited neighbor, if any, of this vertex.
    @return  Either a vertex that is an unvisited neighbor or null
             if no such neighbor exists. */
public VertexInterface<T> getUnvisitedNeighbor();
```

- LISTING 30-1 An interface for the vertices in a graph

Vertices and Edges — Interface (Part 3)

```
 /** Records the previous vertex on a path to this vertex.
    @param predecessor  The vertex previous to this one along a path.  */
  public void setPredecessor(VertexInterface<T> predecessor);

  /** Gets the recorded predecessor of this vertex.
    @return  Either this vertex's predecessor or null if no predecessor
             was recorded. */
  public VertexInterface<T> getPredecessor();

  /** Sees whether a predecessor was recorded for this vertex.
    @return  True if a predecessor was recorded. */
  public boolean hasPredecessor();

  /** Records the cost of a path to this vertex.
    @param newCost  The cost of the path. */
  public void setCost(double newCost);

  /** Gets the recorded cost of the path to this vertex.
    @return  The cost of the path. */
  public double getCost();
} // end VertexInterface
```

- LISTING 30-1 An interface for the vertices in a graph

The Inner Class Edge

```
protected class Edge
{
   private VertexInterface<T> vertex; // Vertex at end of edge
   private double weight;

   protected Edge(VertexInterface<T> endVertex, double edgeWeight)
   {
      vertex = endVertex;
      weight = edgeWeight;
   } // end constructor

   protected Edge(VertexInterface<T> endVertex)
   {
      vertex = endVertex;
      weight = 0;
   } // end constructor

   protected VertexInterface<T> getEndVertex()
   {
      return vertex;
   } // end getEndVertex

   protected double getWeight()
   {
      return weight;
   } // end getWeight
} // end Edge
```

- LISTING 30-2 The protected class `Edge`, as an inner class of `Vertex`

Implementing the Class `Vertex` (Part 1)

```
/** A class of vertices for a graph. */
class Vertex<T> implements VertexInterface<T>
{
   private T label;
   private ListWithIteratorInterface<Edge> edgeList; // Edges to neighbors
   private boolean visited;                          // True if visited
   private VertexInterface<T> previousVertex;        // On path to this vertex
   private double cost;                              // Of path to this vertex

   public Vertex(T vertexLabel)
   {
      label = vertexLabel;
      edgeList = new LinkedListWithIterator<>();
      visited = false;
      previousVertex = null;
      cost = 0;
   } // end constructor

/* Implementations of the vertex operations go here.
   . . .
   */
```

- LISTING 30-3 An outline of the class `Vertex`

Implementing the Class `Vertex` (Part 2)

```
protected class Edge {
   private VertexInterface<T> vertex; // Vertex at end of edge
   private double weight;

   protected Edge(VertexInterface<T> endVertex, double edgeWeight)
   {
      vertex = endVertex;
      weight = edgeWeight;
   } // end constructor

   protected Edge(VertexInterface<T> endVertex)
   {
      vertex = endVertex;
      weight = 0;
   } // end constructor

   protected VertexInterface<T> getEndVertex()
   {
      return vertex;
   } // end getEndVertex

   protected double getWeight()
   {
```

Implementing the Class `Vertex` — method `connect`

```
public boolean connect(VertexInterface<T> endVertex) {
  return connect(endVertex, 0);
} // end connect

public boolean connect(VertexInterface<T> endVertex, double edgeWeight). {
  boolean result = false;

  if (!this.equals(endVertex))
  {  // Vertices are distinct
    Iterator<VertexInterface<T>> neighbors = getNeighborIterator();
    boolean duplicateEdge = false;

    while (!duplicateEdge && neighbors.hasNext())
    {
      VertexInterface<T> nextNeighbor = neighbors.next();
      if (endVertex.equals(nextNeighbor))
        duplicateEdge = true;
    } // end while

    if (!duplicateEdge)
    {
      edgeList.add(new Edge(endVertex, edgeWeight));
      result = true;
    } // end if
  } // end if

  return result;
} // end connect
```

Implementing the Class `Vertex`

- The iterator

```
public Iterator<VertexInterface<T>> getNeighborIterator()
{
   return new NeighborIterator();
} // end getNeighborIterator
```

Implementing the Class Vertex

```
private class NeighborIterator implements Iterator<VertexInterface<T>>. {
 private Iterator<Edge> edges;

 private NeighborIterator()
 {
   edges = edgeList.getIterator();
 } // end default constructor

 public boolean hasNext()
 {
   return edges.hasNext();
 } // end hasNext

 public VertexInterface<T> next()
 {
   VertexInterface<T> nextNeighbor = null;

   if (edges.hasNext())
   {
     Edge edgeToNextNeighbor = edges.next();
     nextNeighbor = edgeToNextNeighbor.getEndVertex();
   }
   else
     throw new NoSuchElementException();

   return nextNeighbor;
 } // end next
```

- LISTING 30-4 The private class `NeighborIterator`, as an inner class of `Vertex`

Implementing the Class `Vertex`

- The methods `hasNeighbor` and `getUnvisitedNeighbor`

```
public boolean hasNeighbor()
{
   return !edgeList.isEmpty();
} // end hasNeighbor

public VertexInterface<T> getUnvisitedNeighbor()
{
   VertexInterface<T> result = null;

   Iterator<VertexInterface<T>> neighbors = getNeighborIterator();
   while ( neighbors.hasNext() && (result == null) )
   {
      VertexInterface<T> nextNeighbor = neighbors.next();
      if (!nextNeighbor.isVisited())
         result = nextNeighbor;
   } // end while

   return result;
} // end getUnvisitedNeighbor
```

Implementing the Class `Vertex`

```
public boolean equals(Object other)
{
   boolean result;
```

- The method `equals`

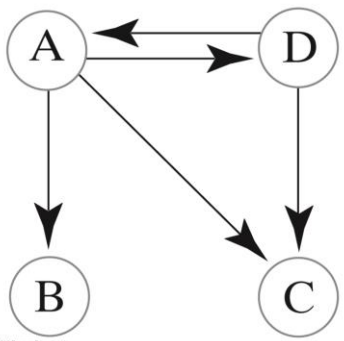
```
   if ((other == null) || (getClass() != other.getClass()))
      result = false;
   else
   {
      // The cast is safe within this else clause
      @SuppressWarnings("unchecked")
      Vertex<T> otherVertex = (Vertex<T>)other;
      result = label.equals(otherVertex.label);
   } // end if

   return result;
} // end equals
```

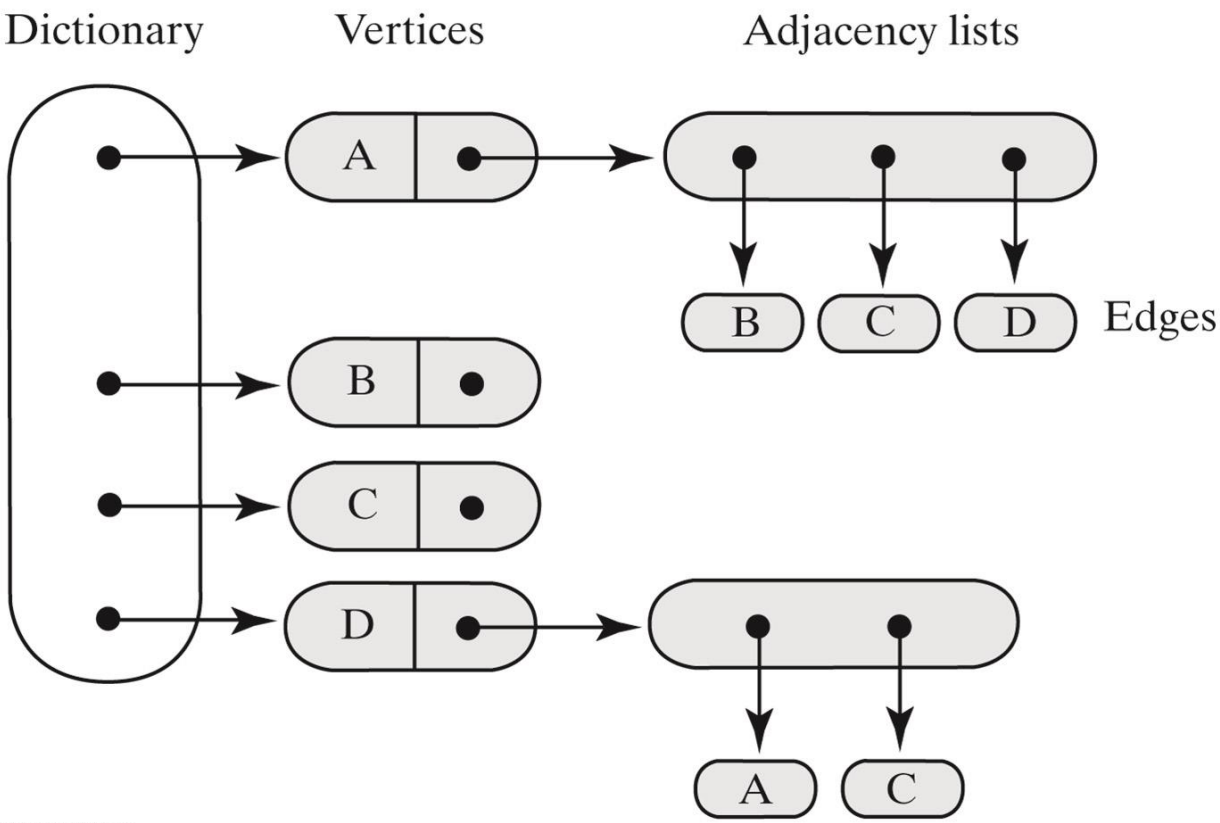
Implementation of the ADT Graph

(a) A directed graph

A rep



(b) An implementation of the graph using a dictionary



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Implementation of the ADT Graph

```
/** A class that implements the ADT directed graph. */
public class DirectedGraph<T> implements GraphInterface<T>
{
    private DictionaryInterface<T, VertexInterface<T>> vertices;
    private int edgeCount;

    public DirectedGraph()
    {
    vertices = new LinkedDictionary<>();
    edgeCount = 0;
    } // end default constructor

/* Implementations of the graph operations go here.
  . . . */

} // end DirectedGraph
```

- LISTING 30-5 An outline of the class `DirectedGraph`

Implementation of the ADT Graph

- The dictionary's method `addVertex`

```
public boolean addVertex(T vertexLabel)
{
   VertexInterface<T> addOutcome =
                 vertices.add(vertexLabel, new Vertex<>(vertexLabel));
   return addOutcome == null; // Was addition to
                                          // dictionary successful?
} // end addVertex
```

Implementation of the ADT Graph

- The `addEdge` methods

```
public boolean addEdge(T begin, T end)
{
   return addEdge(begin, end, 0);
} // end addEdge

public boolean addEdge(T begin, T end, double edgeWeight)
{
   boolean result = false;
   VertexInterface<T> beginVertex = vertices.getValue(begin);
   VertexInterface<T> endVertex = vertices.getValue(end);
   if ( (beginVertex != null) && (endVertex != null) )
      result = beginVertex.connect(endVertex, edgeWeight);
   if (result)
      edgeCount++;
   return result;
} // end addEdge
```

Implementation of the ADT Graph

- Testing for an `edge`

```
public boolean hasEdge(T begin, T end)
{
   boolean found = false;
   VertexInterface<T> beginVertex = vertices.getValue(begin);
   VertexInterface<T> endVertex = vertices.getValue(end);
   if ( (beginVertex != null) && (endVertex != null) )
   {
      Iterator<VertexInterface<T>> neighbors =
                                          beginVertex.getNeighborIterator();
      while (!found && neighbors.hasNext())
      {
         VertexInterface<T> nextNeighbor = neighbors.next();
         if (endVertex.equals(nextNeighbor))
            found = true;
      } // end while
   } // end if
   return found;
} // end hasEdge
```

Implementation of the ADT Graph

- Miscellaneous methods

```
public boolean isEmpty()
{
   return vertices.isEmpty();
} // end isEmpty

public void clear()
{
   vertices.clear();
   edgeCount = 0;
} // end clear

public int getNumberOfVertices()
{
   return vertices.getSize();
} // end getNumberOfVertices

public int getNumberOfEdges()
{
   return edgeCount;
} // end getNumberOfEdges
```

Implementation of the ADT Graph

- Resetting vertices

```
protected void resetVertices()
{
   Iterator<VertexInterface<T>> vertexIterator =
                                                    vertices.getValueIterator();
   while (vertexIterator.hasNext())
   {
      VertexInterface<T> nextVertex = VertexIterator.next();
      nextVertex.unvisit();
      nextVertex.setCost(0);
      nextVertex.setPredecessor(null);
   } // end while
} // end resetVertices
```

Efficiency of Basic Graph Operations

Operation	
addVertex	O(n)
addEdge	O(n)
hasEdge	O(n)
isEmpty	O(1)
getNumberOfVertices	O(1)
getNumberOfEdges	O(1)
Clear	O(1)

- FIGURE 30-4 The performance of basic operations of the ADT graph when implemented by using adjacency lists

Graph Algorithms — Breadth-first traversal

```
public QueueInterface<T> getBreadthFirstTraversal(T origin)
{
  resetVertices();
  QueueInterface<T> traversalOrder = new LinkedQueue<>();
  QueueInterface<VertexInterface<T>> vertexQueue = new LinkedQueue<>();

  VertexInterface<T> originVertex = vertices.getValue(origin);
  originVertex.visit();
  traversalOrder.enqueue(origin);    // Enqueue vertex label
  vertexQueue.enqueue(originVertex); // Enqueue vertex

  while (!vertexQueue.isEmpty())
  {
    VertexInterface<T> frontVertex = vertexQueue.dequeue();
    Iterator<VertexInterface<T>> neighbors = frontVertex.getNeighborIterator();

    while (neighbors.hasNext())
    {
      VertexInterface<T> nextNeighbor = neighbors.next();
      if (!nextNeighbor.isVisited())
      {
        nextNeighbor.visit();
        traversalOrder.enqueue(nextNeighbor.getLabel());
        vertexQueue.enqueue(nextNeighbor);
      } // end if
    } // end while
  } // end while
  return traversalOrder;
} // end getBreadthFirstTraversal
```

Graph Algorithms — Finding the shortest path (Part 1)

```
public int getShortestPath(T begin, T end, StackInterface<T> path)
{
   resetVertices();
   boolean done = false;
   QueueInterface<VertexInterface<T>> vertexQueue = new LinkedQueue<>();
   VertexInterface<T> originVertex = vertices.getValue(begin);
   VertexInterface<T> endVertex = vertices.getValue(end);
   originVertex.visit();

   vertexQueue.enqueue(originVertex);
   while (!done && !vertexQueue.isEmpty())
   {
      VertexInterface<T> frontVertex = vertexQueue.dequeue();
      Iterator<VertexInterface<T>> neighbors = frontVertex.getNeighborIterator();
      while (!done && neighbors.hasNext())
      {
         VertexInterface<T> nextNeighbor = neighbors.next();
         if (!nextNeighbor.isVisited())
         {
            nextNeighbor.visit();
            nextNeighbor.setCost(1 + frontVertex.getCost());
            nextNeighbor.setPredecessor(frontVertex);
```

Graph Algorithms — Finding the shortest path (Part 2)

```
    if (nextNeighbor.equals(endVertex))
        done = true;
  } // end while
} // end while

// Traversal ends; construct shortest path
int pathLength = (int)endVertex.getCost();
path.push(endVertex.getLabel());

VertexInterface<T> vertex = endVertex;
while (vertex.hasPredecessor())
{
  vertex = vertex.getPredecessor();
  path.push(vertex.getLabel());
} // end while

return pathLength;
} // end getShortestPath
```